

KHUTAG-UNDUR, Mongolia

WMO: 442320

Lat: 49.3936N

Lon: 102.7092E

Elev: 3077

StdP: 13.13

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.7	-30.2	-39.7	0.6	-35.5	-34.1	0.9	-29.7	19.2	5.8	16.4	6.4	0.0	210	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	87.7	64.9	83.8	63.2	80.4	62.4	69.0	81.4	66.7	78.3	64.8	75.5	5.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	104.0	74.3	62.9	96.3	71.6	61.0	89.9	69.7	35.2	81.0	33.2	78.6	31.7	75.3	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.3	14.5	-37.3	94.9	6.9	3.8	-42.2	97.7	-46.3	100.0	-50.2	102.1	-55.2	104.9		
(5)			-37.1	72.1	7.1	3.5	-42.2	74.6	-46.3	76.7	-50.3	78.6	-55.4	81.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.2	-10.5	0.8	22.9	41.3	52.8	63.4	67.3	63.1	51.5	35.3	12.8	-4.3
	DBStd	28.56	11.15	12.26	12.30	8.61	8.32	6.85	5.67	5.68	7.48	8.53	12.88	12.72
	HDD50	7772	1875	1377	840	281	68	3	0	1	71	456	1116	1684
	HDD65	11828	2340	1797	1304	712	385	107	39	102	407	920	1566	2149
	CDD50	1639	0	0	0	18	154	405	538	408	115	1	0	0
	CDD65	222	0	0	0	0	6	60	111	44	1	0	0	0
	CDH74	2773	0	0	0	6	248	863	1141	466	49	0	0	0
	CDH80	847	0	0	0	0	59	281	382	121	4	0	0	0
	WSAvg	4.3	1.8	2.6	5.1	6.5	6.4	5.6	5.0	4.8	4.9	4.2	2.9	2.2
	PrecAvg	5.6	0.8	0.7	0.5	0.3	0.4	0.2	0.2	0.2	0.5	0.2	0.5	0.6
(15) Precipitation	PrecMax	15.8	10.7	3.9	4.6	1.2	1.3	1.3	1.0	1.4	2.4	1.1	2.6	9.9
	PrecMin	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.4	1.6	1.3	0.8	0.3	0.4	0.3	0.2	0.3	0.6	0.3	0.7	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.4	36.2	59.9	74.5	86.0	92.6	93.2	89.4	79.3	65.6	48.6	29.4
		MCWB	21.8	28.8	43.4	51.6	58.2	64.4	68.9	67.2	59.8	49.3	38.9	23.7
	2%	DB	17.0	30.3	51.9	67.7	80.7	86.9	88.2	82.8	74.5	59.8	42.0	24.7
		MCWB	13.8	24.6	38.8	48.7	56.7	63.1	66.9	64.1	57.2	46.2	34.0	20.2
	5%	DB	11.7	24.9	46.6	63.1	75.9	82.9	84.1	78.8	70.2	54.9	36.8	20.1
		MCWB	9.6	20.7	36.0	46.3	54.8	61.9	65.3	63.3	55.3	43.2	30.6	16.7
	10%	DB	6.4	19.8	41.5	58.6	71.3	78.9	80.4	75.4	65.8	50.1	31.5	13.9
		MCWB	4.9	16.6	32.8	43.8	52.6	60.5	64.3	62.3	53.2	40.5	27.4	11.8
	0.4%	WB	21.8	29.3	44.6	53.7	62.5	69.0	73.7	70.5	62.7	50.4	38.9	24.2
		MCDB	25.4	35.6	58.8	69.1	79.9	87.1	88.7	82.9	73.8	63.6	47.7	29.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	13.9	24.8	39.6	49.5	59.0	66.3	70.4	67.0	59.1	46.8	34.4	20.3
		MCDB	16.6	30.0	51.0	65.3	76.7	81.8	82.8	78.0	71.5	58.5	41.2	24.5
	5%	WB	9.6	20.8	36.2	46.9	55.9	64.2	68.0	65.0	56.8	43.7	31.1	16.8
		MCDB	11.7	24.7	45.9	61.3	72.7	77.6	79.1	75.1	68.0	54.1	36.4	20.1
	10%	WB	4.8	16.7	33.0	44.3	53.3	62.2	66.0	63.1	54.2	40.8	27.2	11.8
		MCDB	6.3	19.7	40.9	57.8	69.0	74.5	76.1	73.0	64.0	49.2	31.7	13.9
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	26.5	26.9	25.8	26.9	24.7	21.7	22.2	25.6	25.6	20.7	17.6
		MCDBR	24.3	30.2	30.5	33.6	35.8	32.1	29.2	28.8	32.2	32.7	25.5	21.3
		MCWBR	21.7	25.1	19.1	18.9	17.6	13.7	12.0	12.7	15.9	20.0	18.5	17.9
	5% WB	MCDBR	24.2	29.9	29.6	31.2	32.6	27.5	24.4	24.5	29.3	31.6	24.3	21.1
		MCWBR	21.6	24.9	18.8	18.9	17.8	13.6	11.9	12.3	15.2	19.8	18.0	18.0
(40) Clear-Sky Solar Irradiance	taub		0.248	0.272	0.320	0.362	0.393	0.395	0.384	0.359	0.315	0.294	0.270	0.251
	taud		2.322	2.286	2.241	2.247	2.193	2.294	2.376	2.424	2.488	2.468	2.368	2.292
	Ebn at Noon		261	279	281	280	274	274	276	279	283	270	249	240
	Edh at Noon		21	28	36	40	44	40	36	33	28	23	20	19
(44) All-Sky Solar Radiation	RadAvg		534	893	1349	1694	1998	2028	1918	1658	1390	927	562	415
	RadStd		24	48	51	82	103	140	101	95	64	35	25	21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.30	-1.82	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	-3.28	N/A	N/A	-1.31	N/A	N/A	N/A	N/A

Nomenclature: See separate page